# Notes on calculation

Yu Su

*Hefei National Laboratory for Physical Sciences at the Microscale,*
*University of Science and Technology of China*

2026 年 2 月 3 日

# 矩方程

- Hamiltonian

$$H_{\mathrm{T}} = h_{\mathrm{L}} + h_{\mathrm{R}} + \hat{F}_{\mathrm{L}} \hat{F}_{\mathrm{R}} \tag{1}$$

- 耗散子分解

$$\hat{F}_{\alpha=\mathrm{L,R}} = \sum_k \hat{f}_{\alpha k} \tag{2}$$

- 定义

$$f_{\alpha k}(t) \equiv \mathrm{Tr}[\hat{f}_{\alpha k} \rho_{\mathrm{T}}(t)], \quad \sigma_{\alpha k, \alpha' k'}(t) \equiv \mathrm{Tr}[(\hat{f}_{\alpha k} \hat{f}_{\alpha' k'})^{\circ} \rho_{\mathrm{T}}(t)] = \sigma_{\alpha' k', \alpha k}(t) \tag{3}$$

# 矩方程

- 均值的方程

$$\dot{f}_{\mathrm{L}k}(t)=\mathrm{Tr}[\hat{f}_{\mathrm{L}k}\dot{\rho}_{\mathrm{T}}(t)]=-\gamma_{\mathrm{L}k}f_{\mathrm{L}k}(t)-i(\eta_{\mathrm{L}k}-\eta^{*}_{\mathrm{L}\bar{k}})\sum_{k'}f_{\mathrm{R}k'}(t) \tag{4}$$

$$\dot{f}_{\mathrm{R}k}(t)=\mathrm{Tr}[\hat{f}_{\mathrm{R}k}\dot{\rho}_{\mathrm{T}}(t)]=-\gamma_{\mathrm{R}k}f_{\mathrm{R}k}(t)-i(\eta_{\mathrm{R}k}-\eta^{*}_{\mathrm{R}\bar{k}})\sum_{k'}f_{\mathrm{L}k'}(t) \tag{5}$$

由初值 $f_{\alpha k}(0)=0$ 可知

$$f_{\alpha k}(t)=0 \tag{6}$$

# 矩方程

- 方差的方程

$$\dot{\sigma}_{\mathrm{L}k,\mathrm{L}k'} = -(\gamma_{\mathrm{L}k} + \gamma_{\mathrm{L}k'})\sigma_{\mathrm{L}k,\mathrm{L}k'} - i(\eta_{\mathrm{L}k} - \eta^{*}_{\mathrm{L}\bar{k}}) \sum_{j} \sigma_{\mathrm{L}k',\mathrm{R}j}$$
$$- i(\eta_{\mathrm{L}k'} - \eta^{*}_{\mathrm{L}\bar{k}'}) \sum_{j} \sigma_{\mathrm{L}k,\mathrm{R}j} \tag{7}$$

$$\dot{\sigma}_{\mathrm{R}k,\mathrm{R}k'} = -(\gamma_{\mathrm{R}k} + \gamma_{\mathrm{R}k'})\sigma_{\mathrm{R}k,\mathrm{R}k'} - i(\eta_{\mathrm{R}k} - \eta^{*}_{\mathrm{R}\bar{k}}) \sum_{j} \sigma_{\mathrm{R}k',\mathrm{L}j}$$
$$- i(\eta_{\mathrm{R}k'} - \eta^{*}_{\mathrm{R}\bar{k}'}) \sum_{j} \sigma_{\mathrm{R}k,\mathrm{L}j} \tag{8}$$

$$\dot{\sigma}_{\mathrm{L}k,\mathrm{R}k'} = -(\gamma_{\mathrm{L}k} + \gamma_{\mathrm{R}k'})\sigma_{\mathrm{L}k,\mathrm{R}k'} - i(\eta_{\mathrm{L}k}\eta_{\mathrm{R}k'} - \eta^{*}_{\mathrm{L}\bar{k}}\eta^{*}_{\mathrm{R}\bar{k}'})$$
$$- i(\eta_{\mathrm{L}k} - \eta^{*}_{\mathrm{L}\bar{k}}) \sum_{j} \sigma_{\mathrm{R}k',\mathrm{R}j} - i(\eta_{\mathrm{R}k'} - \eta^{*}_{\mathrm{R}\bar{k}'}) \sum_{j} \sigma_{\mathrm{L}k,\mathrm{L}j} \tag{9}$$

# 运动方程

$$\dot{\rho}_{\mathbf{n}}^{(n)}(t) = -\left(H_{\mathrm{S}}^{\times} + \sum_k n_k \gamma_k\right)\rho_{\mathbf{n}}^{(n)}(t) - i\hat{Q}_1 \rho_{\mathbf{n}}^{(n)}(t; \hat{F}_1^{r_1}) - i\hat{Q}_2 \rho_{\mathbf{n}}^{(n)}(t; \hat{F}_2^{r_2}) \cdots$$
$$- i\hat{Q}_{12} \rho_{\mathbf{n}}^{(n)}(t; \hat{F}_1^{s_1} \hat{F}_2^{s_2}) + \cdots \tag{10}$$

$$\hat{F}_a = \sum_k p_{ak} \hat{f}_k \tag{11}$$

$$\rho_{\mathbf{n}}^{(n)}(t; \hat{F}_a) = \sum_k p_{ak} \left[\sqrt{n_k + 1}\sqrt{||\eta_k||}\rho_{\mathbf{n}_k^+}^{(n+1)}(t) + \frac{\sqrt{n_k}\eta_k}{\sqrt{||\eta_k||}}\rho_{\mathbf{n}_k^-}^{(n-1)}(t)\right] \tag{12}$$

$$\rho_{\mathbf{n}}^{(n)}(t; \hat{\Phi}_a) = \sum_k -\gamma_k p_{ak} \left[\sqrt{n_k + 1}\sqrt{||\eta_k||}\rho_{\mathbf{n}_k^+}^{(n+1)}(t) - \frac{\sqrt{n_k}\eta_k}{\sqrt{||\eta_k||}}\rho_{\mathbf{n}_k^-}^{(n-1)}(t)\right] \tag{13}$$

$$\rho_{\mathbf{n}}^{(n)}(t; \hat{F}_a^2) = \sum_k p_{ak} \left[\sqrt{n_k + 1}\sqrt{||\eta_k||}\rho_{\mathbf{n}_k^+}^{(n+1)}(t; \hat{F}_a) + \frac{\sqrt{n_k}\eta_k}{\sqrt{||\eta_k||}}\rho_{\mathbf{n}_k^-}^{(n-1)}(t; \hat{F}_a)\right] \tag{14}$$